Thomas Tony | Curriculum vitae

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Positions

04/2025 – present Doctoral Research Associate, University of Potsdam.
09/2022 – 03/2025 Doctoral Research Associate, University of Münster.

Education

2022 – present PhD student in Mathematics (Dr. rer. nat.), University of Münster.
- Advisor: Rudolf Zeidler

2019 – 2022 Master of Science in Mathematics, University of Freiburg.
2013 – 2019 First State Exam in Mathematics and Physics (teaching degree), University of Freiburg.

Scholarships, grants and awards

2025 Oberwolfach Leibniz Graduate Students travel grant.
2022 Doctoral position funded by the Cluster of Excellence Mathematics Münster.
2020 Deutschlandstipendium, University of Freiburg.
2013 DMV-Abiturpreis, Deutsche Mathematiker-Vereinigung.

Research stays

11/2025 Visiting Rafe Mazzeo (four weeks), Stanford University.

Invitations to conferences and workshops

06/2025 Oberwolfach workshop: Differentialgeometrie im Großen, talk given.
02/2024 Oberwolfach workshop: Analysis, Geometry and Topology of Positive Scalar Curvature Metrics.

Invited research seminar talks

05/2026 Oberseminar Differentialgeometrie, University of Freiburg.
12/2025 Geometry Seminar, Stanford University.
12/2025 Differential Geometry Seminar, UC Santa Barbara.
11/2025 Differential Geometry Seminar, UC Berkeley.
11/2025 Noncommutative Geometry Seminar, Texas A&M University.
11/2025 Geometry Seminar, Texas A&M University.
10/2025 Geometric Analysis Seminar, Columbia University.
07/2025 Oberseminar Topology and Geometry, University of Göttingen.
06/2025 Oberseminar Differentialgeometrie, University of Augsburg.
05/2025 Forschungsseminar Differentialgeometrie, University of Potsdam.
04/2025 Oberseminar Topologie, University of Münster.
04/2024 Arbeitsgruppenseminar Ammann, University of Regensburg.
01/2024 Oberseminar Differentialgeometrie, University of Freiburg.
11/2023 Geometric Analysis Seminar, The City University of New York.
01/2023 Arbeitsgruppenseminar Tuschmann, Karlsruhe Institute of Technology.

Organized conferences and workshops

08/2024 Young Topologist Meeting (YTM), Cluster of Excellence Mathematics Münster.

Selection of conference and workshop participations

06/2026 Geometric Analysis: Parabolic and Elliptic Methods, KTH Stockholm.

10/2025 Geometry: exchanges and perspectives 2025, Institut Henri Poincaré, Paris.

05/2025 Geometric Moduli Spaces - Rigidity, Genericity, Stability, poster presented, ICMS, Edinburgh.

11/2024 Curvature and Geometric Analysis in Rome, poster presented, Sapienza Università di Roma.

12/2023 Workshop on Operator Algebras and Applications: Non-Commutative Geometry, Fields Institute, Toronto.

10/2023 Autumn School on Large Scale Geometry, Gong Talk given, University of Göttingen.

Teaching experience

University of Münster

Winter 2024/25 *Differential Geometry PhD Seminar*, organizer.

Summer 2023 *Differential Equations*, Tutor, Lecture given by Tim de Laat.

University of Freiburg (as a tutor)

2015 – 2022 *Linear Algebra* I and II (2x), *Analysis* I and II, *Multiple Integration* (teaching degree), *Complex Analysis*, *Differential Geometry*.

High School Teaching & Instruction

2016 – 2022 Learn coaching, Das Lernteam.

2016 – 2020 Preparation for the General Certificate of Education in mathematics, Abiturma.

Scientific memberships

2022 – present Cluster of Excellence "Mathematics Münster: Dynamics – Geometry – Structure".

2022 – present CRC 1442 "Geometry: Deformations and Rigidity".

2022 – present Mathematics Münster Graduate School.

Academic services

Reviewer for Science China Mathematics and zbMATH.

Language skills

German (native), English (fluent), French (basic).

Technical skills

Latex, Python, HTML, Linux.

List of publications

Peer-reviewed articles

[1] T. Tony. "Ricci-flat manifolds, parallel spinors and the Rosenberg index." In: *SIGMA Symmetry Integrability Geom. Methods Appl.* 21 (2025), Paper No. 072. DOI: 10.3842/SIGMA.2025.072.

[2] T. Tony. "Scalar curvature rigidity and the higher mapping degree." In: *Journal of Functional Analysis* 288.3 (2025). DOI: 10.1016/j.jfa.2024.110744.

Preprints

[3] O. Riedler and T. Tony. *Scalar-rigid submersions are Riemannian products*. 2026. arXiv: 2601.14117 [math.DG].

Surveys and Reports

[4] T. Tony. "Scalar Curvature Rigidity and the Higher Mapping Degree." In: *Oberwolfach Report No. 30/2025*. Ed. by Richard Bamler et al. 2025. DOI: 10.14760/OWR-2025-30.

Theses

[5] T. Tony. *The End-Periodic Index Theorem for Dirac Operators and its Connection to the Classical Atiyah-Patodi-Singer Index Theorem*. https://ttony.eu/documents/thesis-master.pdf. Master's thesis, advisor: Nadine Große, University of Freiburg. 2022.

[6] T. Tony. *Die Riemannsche Vermutung und die Primzahlverteilung*. https://ttony.eu/documents/thesis-state-exam.pdf. State exam's thesis (accepted as Bachelor's thesis as well), advisor: Katrin Wendland, University of Freiburg. 2019.